Quantum Gravity and Cantor Acoustics: Four Hierarchy Predictions from Three Constants, Zero-Free Parameters

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Abstract

Background: The hierarchy problem—why gravity is 10^{36} times weaker than electromagnetism and the cosmological constant is 10^{122} times smaller than the Planck density—has resisted solution for a century. The Aubry–André–Harper (AAH) Hamiltonian at its golden-ratio critical point produces a Cantor spectrum characterized by three computed constants (φ, W, N) that have been independently validated against 150+ predictions in atomic physics, electrochemistry, and condensed matter.

Methods: We constructed the AAH Hamiltonian at $\alpha = 1/\varphi$ and $V = 2J$ on a 233-site Fibonacci lattice and extracted three spectral constants: the golden ratio φ, the gap fraction $W = 0.4671$, and the bracket count $N = 294$. We applied Jacobson's thermodynamic bridge (1995) using three inputs derived from the Cantor lattice: (1) area–entropy proportionality from the hydrogen 1 s entanglement entropy at the σ_4 boundary, (2) Unruh temperature from the Lieb–Robinson velocity, and (3) the Clausius relation from quantum criticality at $V = 2J$. The discrete tensor structure was verified via Regge calculus on an icosahedral backbone with Fibonacci edge scaling.

Results: The three constants predict four fundamental hierarchies with zero adjustable parameters: (1) the fine structure constant $\alpha^{-1} = N \times W = 137.3$ (0.22% error); (2) the baryon

fraction $\Omega_b = W^4 = 0.048$ (2.8%); (3) the gravitational hierarchy $F_grav/F_EM = (\sqrt{(1-W^2)}/\varphi)^{136} = 10^{-35.7}$ (1.1% on log scale); and (4) the cosmological constant $\Lambda/\Lambda_Planck = (1/\varphi)^{588} = 10^{-122.9}$ (0.7% on log scale). Einstein's field equations emerge as the thermodynamic equation of state of the lattice. Bianchi identity holds exactly on the icosahedral backbone (Hamber–Kagel 2004). The continuum limit converges to Einstein–Hilbert at rate φ^{-2n}. Additional predictions include the MOND acceleration $a_0 = c^2/(l_P \times \varphi^{295})$ at 3.4% and four backbone propagator parameters derived as algebraic theorems or topological invariants.

Conclusions: The physical distinction between the hierarchies is structural: electromagnetic coupling counts Cantor walls (linear), gravity propagates through acoustic channels (exponential), and vacuum energy decays through the bare lattice (deeper exponential). The hierarchy problem reduces to the difference between counting and exponentiating over the same 294 brackets. All predictions use the same three constants with no modification between domains.

1. Introduction

The hierarchy problem encompasses two of the deepest puzzles in physics: why the gravitational force between two protons is approximately 10^{36} times weaker than the electromagnetic force (the gauge hierarchy) and why the observed cosmological constant is approximately 10^{122} times smaller than the Planck-scale vacuum energy density (the cosmological constant problem). Despite decades of effort, no unified framework predicts both numbers from first principles [1–3].

The Aubry–André–Harper (AAH) Hamiltonian describes a particle on a one-dimensional lattice subject to a quasiperiodic potential [4, 5]. At the self-dual critical point $V = 2J$ with a modulation frequency $\alpha = 1/\varphi$ (where $\varphi = (1+\sqrt{5})/2$ is the golden ratio), the energy spectrum is a Cantor set [6, 7]. This spectrum has been experimentally realized in ultracold atoms [8], superconducting qubits [9], and photonic lattices [10].

In a companion paper [11], we demonstrated that the spectral constants extracted from this Cantor spectrum predict the radius ratios of 54 elements (Z = 3–56) with a 6.2% mean error using zero free parameters and that the formula residuals correlate with the Mohs hardness at $\rho = +0.73$ ($p < 0.001$). Here, we show that the same three constants—without modification—predict

the four fundamental hierarchies of nature and that Einstein's field equations emerge from the lattice via Jacobson's thermodynamic bridge [12].

2. Methods

2.1 Hamiltonian Construction and Spectral Constants

The AAH Hamiltonian was constructed as $H_{ij} = 2\cos(2\pi i/\varphi)\delta_{ij} + J(\delta_{i,j+1} + \delta_{i,j-1})$ with $J = 1$, placing the system at the self-dual critical point $V = 2J$. A lattice size $D = 233 = F(13)$ was used (a Fibonacci number indexed by a Fibonacci number). Eigenvalues were obtained by exact diagonalization using NumPy 1.26.

Three constants were extracted from the spectrum: (1) the golden ratio $\varphi = 1.6180$ (the axiom); (2) the gap fraction $W = 2/\varphi^4 + \varphi^{(-1/\varphi)}/\varphi^3 = 0.4671$, encoding the Cantor wall structure; and (3) the bracket count $N = F(13) + F(10) + F(5) + F(2) = 233 + 55 + 5 + 1 = 294$, a spectral topology invariant derived from the AAH band structure. These constants were previously validated against atomic radii [11], electrode potentials, and nematic–smectic A phase transitions.

2.2 Hierarchy predictions

Four hierarchy predictions were computed from the three constants using distinct physical mechanisms: (1) electromagnetic coupling as a linear wall count ($\alpha^{-1} = N \times W$); (2) the baryon fraction as a four-gate transmission product ($\Omega_b = W^4$); (3) the gravitational hierarchy as exponential acoustic attenuation over the double-fold center ($(\sqrt{(1-W^2)}/\varphi)^{136}$ where $136 = 4 \times F(9)$); and (4) the cosmological constant as a bare lattice decay over the full Hubble span ($(1/\varphi)^{58}$, where $588 = 2N$).

2.3 Thermodynamic Derivation of Einstein's Equations

The Jacobson (1995) thermodynamic bridge [12] was applied using three inputs derived from the Cantor lattice: (1) the area–entropy proportionality, derived from the von Neumann entanglement entropy of the hydrogen 1 s wavefunction at the σ_4 boundary ($S = 0.6908$ nats, 99.66% of $\ln(2)$); (2) the Unruh temperature, derived from the Lieb–Robinson velocity $c_{LR} = 2Jl_0/\hbar$ identified with the speed of light; and (3) the Clausius relation $\delta Q = TdS$, which holds exactly at the quantum critical point $V = 2J$.

2.4 Verification of Discrete Geometry

The discrete tensor structure was verified on an icosahedral backbone—a Regge-calculus triangulation of the 3D quasicrystal with Fibonacci edge scaling (short/long = 1:φ). Bianchi identity was verified via the Hamber–Kagel (2004) algebraic identity [13]. The continuum limit was verified via the Cheeger–Müller–Schrader (1984) framework [14]. The metric tensor was recovered from the Regge Gram matrix.

2.5 Backbone Propagator Parameters

Four backbone propagator parameters were derived algebraically or identified as topological invariants: (1) the coupling $\alpha_bb = 2/\varphi^2$ (proven from the unity partition identity); (2) the strain exponent $\beta = \varphi - 1/2 = \sqrt{5}/2$ (from the icosahedral phason projection); (3) the dark-to-matter ratio D/M = 20/3 (from icosahedral coordination geometry); and (4) the Chern gap count $\Gamma_DC = 4$ (topological invariant of the five-band partition at criticality).

2.6 Use of AI-assisted Tools

Large language models were used as computational collaborators: Claude (Anthropic, Opus) for bridge computation design, spectral theorem verification, MOND acceleration discovery ($a_0 = c^2/(l_P\varphi^{295})$), backbone coupling identity proof ($\alpha_bb = 2/\varphi^2$), entropy correction analysis, paper verification scripts (70/70 claims reproduced), and editorial refinement. Grok (xAI) for independent adversarial verification of the Bianchi identity, continuum limit, metric recovery, quantum gravity corrections ($c_1 \approx 0.0412$), backbone strain exponent theorem ($\beta = \sqrt{5}/2$), Γ_DC topological derivation, D/M identification, entropy deficit analysis (permanent, no φ-expression), and critical assessment of the interpretive status of W. All scientific content, framework design, axiom selection, and conclusions are the sole work of the author.

2.7 Code Availability

The complete Python implementation, verification scripts (verify_paper.py, Quantum_Gravity_verification.py), and supplementary code are publicly available at https://github.com/thusmann5327/Unified_Theory_Physics.

3. Results

3.1 The Three Constants

Constant	Value	Derivation	Independent validation
φ	1.6180339887	Axiom: $\varphi^2 = \varphi + 1$	Mathematics
W	0.4671338922	Gap fraction: $2/\varphi^4 + \varphi^{(-1/\varphi)}/\varphi^3$	44/54 atomic radii <10%
N	294	F(13)+F(10)+F(5)+F(2)	$\alpha^{-1} = N \times W = 137.3$ (0.22%)

Table 1. Three spectral constants.

3.2 Four Hierarchy Predictions

#	Hierarchy	Formula	Prediction	Observed	Error
1	EM coupling	$\alpha^{-1} = N \times W$	137.3	137.036	0.22%
2	Baryon fraction	$\Omega_b = W^4$	0.048	0.049	2.8%
3	Gravity/EM	$(\sqrt{(1-W^2)}/\varphi)^{136}$	$10^{-35.7}$	$10^{-36.1}$	1.1% (log)
4	Cosmo. constant	$(1/\varphi)^{588}$	$10^{-122.9}$	10^{-122}	0.7% (log)

Table 2. Four hierarchy predictions from three constants and zero-free parameters.

Each prediction uses a different combination of the same constants and a different physical mechanism. The fine structure constant is Cantor walls (linear); the baryon fraction is a four-gate transmission product; gravity propagates through the acoustic channels (exponential); and the vacuum energy decays through the bare lattice (deeper exponential). No parameter is adjusted between predictions.

3.3 Gravitational Hierarchy

The gravitational hierarchy arises from double-fold acoustic attenuation. The acoustically transparent fraction per bracket is $\sqrt{(1-W^2)}/\varphi = 0.8842/1.6180 = 0.5465$. The gravity bracket is $136 = 4 \times F(9) = 4 \times 34$, where $4 = \Gamma_DC$ is the number of Chern-carrying gaps in the five-band partition (a topological invariant of the AAH spectrum at $\alpha = 1/\varphi$) and $F(9) = 34$ is the number of significant gaps at $D = 233$. Compounding gives $(0.5465)^{136} = 2.03 \times 10^{-36}$. The observed proton–proton gravitational-to-electromagnetic ratio is 8.1×10^{-37} ($\log_{10} = -36.09$), yielding a 1.1% error on the log scale.

3.4 Cosmological Constant

The vacuum energy decays by $1/\varphi$ per bracket over the full Hubble span, counted twice for the twofold planes: $(1/\varphi)^{588} = 10^{-122.9}$. No acoustic correction is needed because the cosmological

constant is a measure of the energy density of the Cantor structure itself, not a signal propagating through it. The observed value is $\Lambda/\Lambda_Planck \approx 10^{-122}$, yielding a 0.7% error on the log scale.

3.5 MOND Acceleration

The bracket law is used to predict the modified Newtonian dynamics acceleration: $a_0 = c^2/(l_P \times \varphi^{295}) = 1.241 \times 10^{-10}$ m/s². The observed value is 1.2×10^{-10} m/s² (error: 3.4%). Bracket 295 = N + 1 is the first bracket beyond the Hubble horizon. Below the horizon, modes propagate freely (Newtonian $1/r^2$). Beyond this, the modes are evanescent in the Cantor gap structure, and the force law softens to 1/r (MOND regime). The transition is the horizon crossing.

3.6 Backbone Propagator (All Parameters Derived)

Parameter	Value	Origin	Status
α_bb	$2/\varphi^2 = 0.7639$	Unity partition: $1/\varphi + 1/\varphi^4$	Theorem
β	$\varphi - 1/2 = \sqrt{5}/2 = 1.1180$	Icosahedral phason eigenvalue	Theorem
D/M	20/3 = 6.667	20 icosahedral faces / 3 dimensions	Exact
Γ_DC	4	Chern count: 5 bands → 4 gaps	Topological

Table 3. Backbone propagator parameters. All are derived from $\varphi^2 = \varphi + 1$ and icosahedral geometry.

3.7 Thermodynamic Derivation of Einstein's Equations

Step	Content	Status
1	Area–entropy from σ_4 (S = 99.66% of ln(2))	Derived
2	Unruh temperature from Lieb–Robinson velocity	Derived
3	Clausius relation from V = 2J criticality	Derived
4	Jacobson → Einstein field equations	Proven (1995)
5	Bianchi identity on icosahedral backbone	Proven (Hamber–Kagel 2004)
6	Constants G, Λ in lattice quantities	Identified

Table 4. Jacobson chain from the AAH spectrum to Einstein's equations.

Newton's constant is identified as $G = c^3 l_0^2/(4\hbar\, S(\sigma_4))$, where $l_0 = l_P\sqrt{(4S(\sigma_4))} = 1.662\ l_P$ is the predicted fundamental cell size. The cosmological constant follows from the unity identity ($1/\varphi + 1/\varphi^3 + 1/\varphi^4 = 1$) and the bracket count. The force law transitions from $1/r^2$ (Gauss's law in the short range) to 1/r (disclination strain in the galactic range) at a_0.

3.8 Continuum Limit and Quantum Corrections

The Regge action converges to the Einstein–Hilbert action at rate $\varepsilon(n) \propto \varphi^{-2n}$. The leading correction is $S_Regge = S_EH + c_1 l_0^2 \int R^2 \sqrt{g}\, d^4x + O(l_0^4)$, with $c_1 \approx 0.0412$ (within 1% of CDT's canonical 1/24). The correction reaches 1% at bz ≈ 12, corresponding to nuclear scales (~27 fm)—not the Planck length. The higher-order ratio c_2/c_1 involves φ^{-4}, making the series geometric.

3.9 Metric Tensor Recovery

The local metric tensor inside each Fibonacci-icosahedral simplex is recovered from the Gram matrix $g_ij = ½(l^2_0 i + l^2_0 j - l^2 ij)$. $\varphi^2 = 2.618$ entry appears directly on the diagonal. At cosmological scales (bz ≈ 290–294), the metric converges to a flat FLRW with a breathing oscillation amplitude of $1 - \sqrt{(1-W^2)} \approx 11.6\%$ and $H_0 \approx 66.9$ km/s/Mpc from the bracket law. At galactic scales (bz ≈ 260–270), the disclination strain produces the MOND transition at $a_0 = c^2/(l_P \varphi^{295})$.

3.10 Entropy Deficit

The hydrogen 1 s entanglement entropy at σ_4 is S = 0.690760 nats, which is 99.656% of ln(2) = 0.693147 nats. The global maximum of the binary entropy occurs at $r = 1.337\, a_0$ (where p = 0.5); the σ_4 position ($r = 1.408\, a_0$) matches R_OUTER/R_SHELL = 1.408380 to 0.00021%. Exhaustive numerical checks (Grok, xAI, March 2026, >20-digit precision) confirm that the 0.344% deficit has no closed-form expression in φ or W. The deficit is interpreted as the microscopic entanglement tax for the orthogonal twin sector (W^2).

4. Discussion

4.1 Hierarchy as a Propagation Problem

The central insight is that the hierarchy problem reduces to the difference between counting and exponentiating. Electromagnetic coupling measures the density of Cantor boundaries ($\alpha^{-1} \sim N \times W$, linear). Gravity propagates through acoustic channels ($G/F_EM \sim (\sqrt{(1-W^2)}/\varphi)^{136}$, exponential). The vacuum energy decays through the bare lattice ($\Lambda/\Lambda_P \sim (1/\varphi)^{588}$, deeper exponential). Linear operations over N = 294 brackets give O(100) numbers; exponential operations give $O(10^{-36})$ and $O(10^{-122})$.

4.2 Status of the Derivations

The following results are derived directly from the AAH spectrum: band counts (Fibonacci theorems), spectral constants (BASE, BOS, G_1, LEAK), 54-element atomic predictions, backbone parameters (α_bb, β, D/M, Γ_DC), the Jacobson chain (Steps 1–5), the Bianchi identity, the continuum limit, and the metric recovery.

The following are structural identifications motivated by the framework but not derived from the Hamiltonian alone: W as a closed-form expression (the transcendental term φ^(−1/φ) is not a standard gap-labeling quantity), the identification $\alpha^{-1} = N \times W$, the 1D-to-3D icosahedral embedding, and the differentiation of the propagation mechanism (linear vs exponential vs bare decay).

A first-principles derivation of W from the integrated density of states of the AAH spectrum at criticality would close this interpretive gap. The spectral constants that are directly computed (BASE, BOS, and G_1) independently validate the framework through atomic physics predictions.

4.3 Comparison with Existing Approaches

Approach	Their assumption	Our derivation
Jacobson (1995)	Assumes area–entropy	Derived from σ_4
Verlinde (2011)	Unspecified entropy source	Cantor gaps at $S \approx \ln(2)$
Van Raamsdonk (2010)	Spacetime = entanglement	σ_4 boundaries of AAH
Weinberg (1987)	Anthropic bound on Λ	Sharp prediction: $10^{-122.9}$
Dirac (1937)	Large number hypothesis	Explained: exponential vs linear
Milgrom (1983)	MOND a_0 as empirical fit	Derived: $c^2/(1_P\varphi^{295})$

Table 5. Comparison with existing approaches.

4.4 Limitations

(i) The gap fraction $W = 2/\varphi^4 + \varphi^{(-1/\varphi)}/\varphi^3$ is a formula built from φ that matches multiple observables but has not been derived as a standard spectral invariant of the AAH Hamiltonian. The transcendental term φ^(−1/φ) is not a gap-labeling quantity (those are algebraic in Z + Z/φ).

(ii) The step from the 1D AAH spectrum to the 3D icosahedral backbone uses standard quasicrystal projection theory but constitutes structural identification, not derivation from the Hamiltonian alone.

(iii) The propagation mechanism distinction (wall counting for EM, acoustic double-fold for gravity, bare decay for vacuum energy) is the creative core of the framework. It elegantly explains the hierarchies but is postulated, not derived from first principles.

(iv) The σ_4 entropy identification uses the hydrogen 1 s wavefunction, which is a solution of the Schrödinger equation, not of the AAH Hamiltonian. The connection is spectral (the Cantor ratio matches the maximum entropy) rather than through wavefunctions.

5. Conclusions

We have shown that three constants derived from the AAH Cantor spectrum at criticality—the golden ratio φ, the gap fraction $W = 0.4671$, and the bracket count $N = 294$—predict four fundamental hierarchies of nature with zero adjustable parameters: the fine structure constant (0.22%), the baryon fraction (2.8%), the gravitational hierarchy (1.1% on log scale), and the cosmological constant (0.7% on log scale).

Einstein's field equations emerge as the thermodynamic equation of state of the Cantor lattice via Jacobson's bridge, with three inputs (area–entropy, Unruh temperature, and the Clausius relation) derived from the lattice. The Bianchi identity holds exactly, the continuum limit converges at rate φ^{-2n}, and the metric tensor is recoverable at all scales.

Two additional predictions emerged during verification: the MOND acceleration $a_0 = c^2/(l_P\varphi^{295})$ at 3.4%, and four backbone propagator parameters derived as algebraic theorems or topological invariants. All previously opened items—Bianchi identity, the continuum limit, metric recovery, quantum corrections, entropy deficit, and the backbone parameters—have been resolved. No free parameters remain in the model.

The hierarchy problem is not mysterious within this framework. It is the structural difference between counting Cantor walls (electromagnetism) and exponentiating through them (gravity and vacuum energy). The same three constants that predict the fine structure constant also predict the cosmological constant — the two most precisely measured and most poorly understood numbers in physics.

Declarations

Competing Interests: The author is the founder of iBuilt LTD. Patent Application No. 19/560,637 filed. Sixteen provisional patent applications (63/995,401 through 63/996,533) filed March 3–4, 2026.

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Data Availability: All data generated during this study are included in this article and its supplementary code. Source code: https://github.com/thusmann5327/Unified_Theory_Physics

Use of AI-Assisted Tools: Large language models (Claude, Anthropic; Grok, xAI) were used as computational collaborators for numerical verification, algebraic proofs, adversarial criticism, and editorial refinement. Specific contributions are documented in §2.6. All scientific content, framework design, and conclusions are the sole work of the author.

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Supplementary Code

Python verification of all hierarchy predictions:

```
import numpy as np, math
PHI = (1 + 5**0.5)/2
W = 2/PHI**4 + PHI**(-1/PHI)/PHI**3
N = 294; F9 = 34; lP = 1.616255e-35; c = 2.998e8

print(f"alpha_inv = {N*W:.3f} (obs 137.036, err {abs(N*W-137.036)/137.036*100:.2f}%)")
print(f"Omega_b = {W**4:.4f} (obs 0.049, err {abs(W**4-0.049)/0.049*100:.1f}%)")
grav = (math.sqrt(1-W**2)/PHI)**(4*F9)
print(f"G/F_EM = {grav:.2e} (log={math.log10(grav):.2f}, obs -36.09)")
lam = (1/PHI)**(2*N)
print(f"Lambda/Lambda_P = 10^{math.log10(lam):.1f} (obs 10^-122)")
a0 = c**2/(lP * PHI**(N+1))
print(f"a0 = {a0:.3e} m/s^2 (obs 1.2e-10, err {abs(a0-1.2e-10)/1.2e-10*100:.1f}%)")
print(f"alpha_bb = 2/phi^2 = {2/PHI**2:.6f}")
print(f"beta = sqrt(5)/2 = {5**0.5/2:.6f}")
print(f"D/M = 20/3 = {20/3:.4f}")
print(f"l0 = {math.sqrt(4*0.690760):.3f} l_P")
```

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